Edward Chen

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Projects

KV-Cache Compression for Transformer Inference | **local-turboquant** | PyTorch, Modal, transformers, CUDA

- Reproduced an unpublished ICLR 2026 paper from scratch and shipped it as a working library before the authors released code. Shipped with one-line API, wrapper and a CLI exposing per-generation savings, throughput, and freed VRAM; benchmarked on H100s via Modal.
- Fused Triton attention kernel that runs directly on bit-packed 3-bit KV indices without decompressing to FP16, achieving 74% KV cache memory reduction at near-lossless quality

Modified Transformer Architecture | smarternano | PyTorch, CUDA, OpenRouter, transformers

- Re-engineered the nanochat architecture with custom layer depth and weight initialization, re-trained on Nvidia's Nemotron Nano dataset, publicly recognized by Andrej Karpathy.
- Built a synthetic data generation pipeline via OpenRouter to align the model's personality, achieving coherence comparable to 560M parameter models.

AI-Native App | murmur | ONNX, Flutter, AI Engineering

- Built and released a mobile app that converts any PDF into human-quality audio entirely on-device, achieving 3-second load-to-playback with zero server dependencies.Deployed Kokoro-82M transformer TTS on Android via ONNX Runtime, forking the runtime to isolate inference on a background thread for performance.
- Built an NLP preprocessing pipeline by cross-compiling espeak-ng and writing Dart FFI bindings for IPA phonemization, implementing token-aware batch splitting against the model's tokenizer vocab, and caching per-sentence phonemes in SQLite to eliminate redundant compute at inference time.

Education

Arizona State University August 2023 - Present

Major: BS Data Science, BA Supply Chain Management, Minor in Economics (GPA: 3.7)
Coursework: Data Structures, Operating Systems, Computer Vision, Deep Learning, Machine Learning, Linear Algebra
Activities: Scholars of Finance, Association of Computing Machinery

Experience

Age of Learning | Data Engineer Intern, AI/ML Systems June 2026 - Present

- Built a production agentic AI monitoring pipeline for 6 core stakeholder-facing Omni reports, using Snowflake Cortex, custom tool-calling agents, and directed post-training to identify anomalous report runs with 90% recall.
- Extended report-specific agents to investigate related GitLab merge requests, map anomalous metrics to likely code owners, and route incident summaries to engineering teams through a custom Slack MCP workflow.
- Engineered a next-activity recommendation system for ranking personalized learning activities after lesson completion, improving learner continuation rate by 20% while preserving learning requirements.

mymelo.org | Co-Founder, ML Engineer April 2025 - June 2026

- Won Caltech HackTech 2025. Trained EfficientNet-B0 on 30K dermoscopy images with transfer learning and class-weighted augmentation, achieving 95% recall for melanoma triage across clinical deployment.
- Engineered AWS S3 + Lambda pipeline overnight to compress 3-5MB images to 300KB WebP, reducing per-image disk footprint by 94% while preserving originals in S3 for compliance. Built mobile platform end-to-end.

M.Y Intellectual Property | Software Engineer Intern May 2024 - August 2024

- Engineered a unified patent search API aggregating USPTO, CNIPA, and EPO databases into a single query interface, reducing cross-jurisdictional research from 3 workflows to 1 and cutting attorney lookup time by 60%.
- Built citation network parser to extract patent family relationships across 50K+ international filings, automating competitor portfolio analysis for 15+ client engagements.

Skills

Languages: C++, Python, Java, CUDA, Rust
Technologies: AWS, Docker, Flutter, React, Next.js, Django, Flask, Supabase, Git, MongoDB, Snowflake, SQLite, Modal
AI/ML: PyTorch, FastAPI, ONNX Runtime, OpenCV, vLLM, Post Training, FAISS, ANOVA, Transformers, JAX
Contributions: Meta PyTorch